Mathematics for Electrical Engineering and Signal Processing

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1 The Quadratic Formula

For $ax^2 + bx + c = 0$ with $a \neq 0$, the solutions are

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

The quantity $D = b^2 - 4ac$ is the *discriminant*. There are two distinct real roots when $D > 0$, one repeated real root when $D = 0$, and two complex conjugate roots when $D < 0$. Here and throughout this review, $\mathrm{j} = \sqrt{-1}$, so $\sqrt{D} = \mathrm{j}\sqrt{-D}$ when $D < 0$. To see where the formula comes from, completing the square gives

$$(2ax + b)^2 = b^2 - 4ac.$$

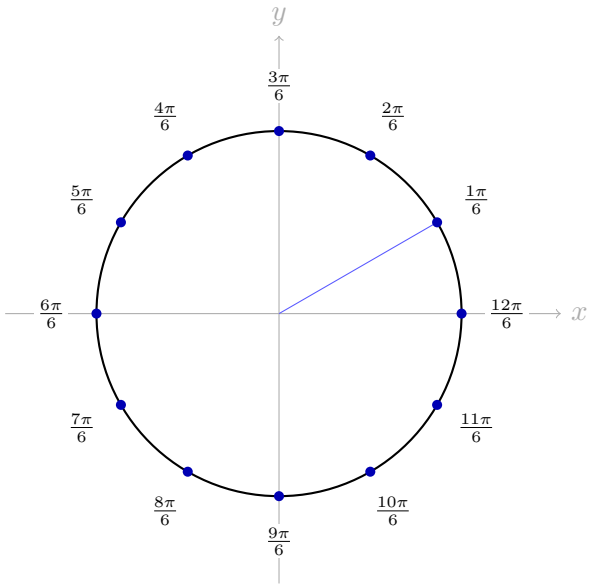
Taking both square roots and solving for x yields the formula above.

Exercises

1. Solve $x^2 - 5x + 6 = 0$.
2. Solve $x^2 + 4x + 4 = 0$.
3. Solve $x^2 + 2x + 5 = 0$, using j for the imaginary unit.
4. Solve $2x^2 - 3x - 2 = 0$.

2 Trigonometric Reference Values

On the unit circle, the point at angle θ has coordinates $(\cos\theta, \sin\theta)$. The marked points below are $\theta = k\pi/6$ for $k = 1, 2, \ldots, 12$, measured counterclockwise from the positive horizontal axis.



The following table lists the values of $\sin(k\pi/6)$ and $\cos(k\pi/6)$ for $k = 1, 2, \ldots, 12$.

k	1	2	3	4	5	6	7	8	9	10	11	12
$\sin(k\pi/6)$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$	-1	$-\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$	0
$\cos(k\pi/6)$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$	-1	$-\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$	0	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	1

3 Basic Complex Numbers

A complex number has the form $z = a + b\mathrm{j}$, where a and b are real and $\mathrm{j}^2 = -1$. This expression is called the *rectangular form* of z: the real number a is its horizontal (real) component, and the real number b is its vertical (imaginary) component. Thus its real and imaginary parts are $\operatorname{Re} z = a$ and $\operatorname{Im} z = b$. Arithmetic follows the ordinary distributive laws, followed by replacing j^2 by -1.

The conjugate of $z = a + b\mathrm{j}$ is $\overline{z} = a - b\mathrm{j}$. The magnitude is

$$|z| = \sqrt{a^2 + b^2}.$$

Conjugate–magnitude theorem

For every complex number z,

$$z\overline{z} = |z|^2.$$

Derivation. If $z = a + b\mathrm{j}$, then $\overline{z} = a - b\mathrm{j}$, so

$$z\overline{z} = (a + b\mathrm{j})(a - b\mathrm{j}) = a^2 - b^2\mathrm{j}^2 = a^2 + b^2 = |z|^2.$$

To divide by a nonzero complex number, multiply numerator and denominator by the conjugate of the denominator. For example,

$$\begin{aligned} \frac{3 + 2\mathrm{j}}{1 - 2\mathrm{j}} &= \frac{3 + 2\mathrm{j}}{1 - 2\mathrm{j}}\,\frac{1 + 2\mathrm{j}}{1 + 2\mathrm{j}} \\ &= \frac{(3 + 2\mathrm{j})(1 + 2\mathrm{j})}{(1 - 2\mathrm{j})(1 + 2\mathrm{j})} \\ &= \frac{-1 + 8\mathrm{j}}{5} = -\frac{1}{5} + \frac{8}{5}\mathrm{j}. \end{aligned}$$

The result is again in rectangular form.

Polar form

A nonzero complex number $z = a + b\mathrm{j}$ can also be written in *polar form* as

$$z = r(\cos\theta + \mathrm{j}\sin\theta) = r\mathrm{e}^{\mathrm{j}\theta},$$

where $r = |z| = \sqrt{a^2 + b^2}$ and θ is an angle in the correct quadrant satisfying $\cos\theta = a/r$ and $\sin\theta = b/r$. When $a \neq 0$, $\tan\theta = b/a$, but the signs of a and b must be used to choose the correct quadrant.

For example, for $z = 3 + 4\mathrm{j}$,

$$r = \sqrt{3^2 + 4^2} = 5, \qquad \theta = \arctan\left(\frac{4}{3}\right),$$

and the point lies in the first quadrant. Therefore

$$3 + 4\mathrm{j} = 5\left(\cos\left(\arctan\frac{4}{3}\right) + \mathrm{j}\sin\left(\arctan\frac{4}{3}\right)\right) = 5\mathrm{e}^{\mathrm{j}\arctan(4/3)}.$$

Exercises

1. Write $3 - 7\mathrm{j}$ in the form $a + b\mathrm{j}$ and identify its real and imaginary parts.
2. Compute $(2 + 3\mathrm{j}) + (5 - 4\mathrm{j})$.
3. Compute $(2 + 3\mathrm{j})(5 - 4\mathrm{j})$.
4. Compute $(1 + \mathrm{j})^2$.
5. Find $\overline{z}$ when $z = -4 + 9\mathrm{j}$.
6. Find $|3 - 4\mathrm{j}|$.
7. Simplify $\dfrac{1}{1 - \mathrm{j}}$ to rectangular form.
8. Simplify $\dfrac{3 + 2\mathrm{j}}{1 - 2\mathrm{j}}$ to rectangular form.
9. Verify directly that $(2 - 5\mathrm{j})\overline{(2 - 5\mathrm{j})} = |2 - 5\mathrm{j}|^2$.
10. Find all real a, b such that $(a + b\mathrm{j})(1 + \mathrm{j}) = 3 - \mathrm{j}$.
11. Write $1 + \sqrt{3}\,\mathrm{j}$ in polar exponential form.
12. Write $-1 + \mathrm{j}$ in polar exponential form, using an argument between 0 and 2π.

3.1 Arguments, Conversion, and Multiplication in Polar Form

For $z \neq 0$, an *argument* is any angle θ such that $z = |z|\mathrm{e}^{\mathrm{j}\theta}$. Arguments differ by integer multiples of 2π. Here the function $\arg z$ means the *principal argument*, chosen in $(-\pi, \pi]$. Thus all arguments are $\arg z + 2k\pi$, $k \in \mathbb{Z}$. The argument of zero is undefined.

To convert from rectangular to polar form, compute

$$r = \sqrt{a^2 + b^2}, \qquad \theta = \operatorname{atan2}(b, a), \qquad a + b\mathrm{j} = r\mathrm{e}^{\mathrm{j}\theta}.$$

The two-input function atan2 uses both signs to choose the quadrant (with the negative real axis assigned angle π here). Using $\arctan(b/a)$ alone loses quadrant information and fails when $a = 0$. For instance, $-\sqrt{3} + \mathrm{j} = 2\mathrm{e}^{5\mathrm{j}\pi/6}$. To convert back, use Euler's formula:

$$a = r\cos\theta, \qquad b = r\sin\theta, \qquad r\mathrm{e}^{\mathrm{j}\theta} = r\cos\theta + \mathrm{j}r\sin\theta.$$

For example, $4\mathrm{e}^{-\mathrm{j}\pi/3} = 2 - 2\sqrt{3}\mathrm{j}$.

Multiplication is especially simple in polar form:

$$(r_1\mathrm{e}^{\mathrm{j}\theta_1})(r_2\mathrm{e}^{\mathrm{j}\theta_2}) = r_1 r_2 \mathrm{e}^{\mathrm{j}(\theta_1 + \theta_2)}.$$

Multiply the magnitudes and add the angles, reducing the resulting angle modulo 2π if a principal argument is required. For example,

$$(2\mathrm{e}^{\mathrm{j}\pi/3})(3\mathrm{e}^{-\mathrm{j}\pi/6}) = 6\mathrm{e}^{\mathrm{j}\pi/6} = 3\sqrt{3} + 3\mathrm{j}.$$

Since $\mathrm{j} = \mathrm{e}^{\mathrm{j}\pi/2}$, multiplying by j rotates a complex number counterclockwise by 90° without changing its magnitude: $\mathrm{j}(a + b\mathrm{j}) = -b + a\mathrm{j}$. Since $1/\mathrm{j} = -\mathrm{j} = \mathrm{e}^{-\mathrm{j}\pi/2}$, multiplying by $1/\mathrm{j}$ rotates clockwise by 90°: $-\mathrm{j}(a + b\mathrm{j}) = b - a\mathrm{j}$.

Exercises: Arguments and Polar Multiplication

Use the principal-argument convention above unless otherwise stated.

1. Find $\arg(-1-\mathrm{j})$ and list all arguments of $-1-\mathrm{j}$.
2. Find $\arg(-3)$ and $\arg(2\mathrm{j})$. Explain why $\arg(0)$ is undefined.
3. Convert $\sqrt{3}-\mathrm{j}$ to polar exponential form.
4. Convert $-3+3\mathrm{j}$ to polar exponential form.
5. Convert $5\mathrm{e}^{\mathrm{j}\pi/2}$ to rectangular form.
6. Convert $2\mathrm{e}^{-2\mathrm{j}\pi/3}$ to rectangular form.
7. Use polar form to compute $(2\mathrm{e}^{\mathrm{j}\pi/4})(3\mathrm{e}^{\mathrm{j}\pi/6})$; give its magnitude and principal argument.
8. Use polar form to compute $(1+\mathrm{j})(1-\sqrt{3}\mathrm{j})$ in polar and rectangular forms.
9. For $z=2-3\mathrm{j}$, compute $\mathrm{j}z$ and z/j, and describe the rotations.
10. Use polar form to compute $(\sqrt{3}+\mathrm{j})^3$ in rectangular form.

4 Euler's Formula

Euler's formula connects exponentials, trigonometric functions, and complex numbers:

$$\mathrm{e}^{\mathrm{j}\theta}=\cos\theta+\mathrm{j}\sin\theta.$$

Consequently,

$$\cos\theta=\frac{\mathrm{e}^{\mathrm{j}\theta}+\mathrm{e}^{-\mathrm{j}\theta}}{2},\qquad \sin\theta=\frac{\mathrm{e}^{\mathrm{j}\theta}-\mathrm{e}^{-\mathrm{j}\theta}}{2\mathrm{j}}.$$

If $z=r(\cos\theta+\mathrm{j}\sin\theta)$, then $z=r\mathrm{e}^{\mathrm{j}\theta}$; this is the polar form of z. Here $r=|z|$ and θ is an argument of z.

Exercises

1. Use Euler's formula to write $\mathrm{e}^{\mathrm{j}\pi/2}$ in rectangular form.
2. Use Euler's formula to write $\mathrm{e}^{\mathrm{j}\pi}$ in rectangular form.
3. Write $1+\mathrm{j}$ in polar form.
4. Write $-1+\sqrt{3}\mathrm{j}$ in polar form.
5. Convert $4\mathrm{e}^{-\mathrm{j}\pi/3}$ to rectangular form.
6. Compute $(\mathrm{e}^{\mathrm{j}\pi/4})^6$ in rectangular form.
7. Use Euler's formula to simplify $\mathrm{e}^{\mathrm{j}a}\mathrm{e}^{\mathrm{j}b}$.
8. Find the magnitude and one argument of $3\mathrm{e}^{\mathrm{j}\pi/6}$.

5 Trigonometric Addition Formulas

Euler's formula gives a quick derivation. On one hand,

$$e^{j(a+b)} = \cos(a+b) + j\sin(a+b).$$

On the other hand,

$$\begin{aligned} e^{j(a+b)} &= e^{ja}e^{jb} \\ &= (\cos a + j\sin a)(\cos b + j\sin b) \\ &= (\cos a\cos b - \sin a\sin b) + j(\sin a\cos b + \cos a\sin b). \end{aligned}$$

Equating real and imaginary parts yields

$$\begin{aligned} \cos(a+b) &= \cos a\cos b - \sin a\sin b, \\ \sin(a+b) &= \sin a\cos b + \cos a\sin b. \end{aligned}$$

Replacing b by $-b$ gives

$$\begin{aligned} \cos(a-b) &= \cos a\cos b + \sin a\sin b, \\ \sin(a-b) &= \sin a\cos b - \cos a\sin b. \end{aligned}$$

Exercises

1. Use $\sin(a+b)$ to find $\sin(\pi/12)$. *Hint:* $\pi/3 - \pi/4 = \pi/12$.
2. Use $\cos(a-b)$ to find $\cos(\pi/12)$.
3. Expand $\sin(x+\pi/6)$.
4. Expand $\cos(x-\pi/3)$.
5. Simplify $\sin(x+y) + \sin(x-y)$.
6. Simplify $\cos(x+y) + \cos(x-y)$.
7. Verify $\sin(a-b)$ by replacing b with $-b$ in $\sin(a+b)$.
8. Find $\sin(75°)$ using an addition formula.
9. Find $\cos(105°)$ using an addition formula.
10. If $\sin a = 3/5$, $\cos a = 4/5$, $\sin b = 5/13$, and $\cos b = 12/13$, find $\sin(a+b)$.

6 Double-Angle Formulas

Set $b = a$ in the addition formulas. Then

$$\sin(2a) = \sin(a+a) = 2\sin a\cos a.$$

Similarly,

$$\cos(2a) = \cos(a+a) = \cos^2 a - \sin^2 a.$$

Using $\sin^2 a + \cos^2 a = 1$ gives the equivalent forms

$$\cos(2a) = 2\cos^2 a - 1 = 1 - 2\sin^2 a.$$

Exercises

1. Derive $\sin(2a)$ from $\sin(a+b)$.
2. Derive $\cos(2a)$ from $\cos(a+b)$.
3. Rewrite $\cos(2x)$ using only $\sin x$.
4. Rewrite $\cos(2x)$ using only $\cos x$.
5. Simplify $2\sin x\cos x$.
6. Simplify $\cos^2 x - \sin^2 x$.
7. Find $\sin(2\pi/3)$ using a double-angle formula.
8. If $\sin x = 3/5$ and x is in the first quadrant, find $\cos(2x)$.
9. If $\cos x = 5/13$ and x is in the first quadrant, find $\sin(2x)$.
10. Solve $2\sin x\cos x = 1$ for $0 \le x < 2\pi$.

7 Half-Angle Formulas

The double-angle formulas supply the derivation. With $a = \theta/2$,

$$\cos\theta = 1 - 2\sin^2(\theta/2),$$

so

$$\sin^2(\theta/2) = \frac{1-\cos\theta}{2}.$$

Likewise, from $\cos\theta = 2\cos^2(\theta/2) - 1$,

$$\cos^2(\theta/2) = \frac{1+\cos\theta}{2}.$$

Thus

$$\sin(\theta/2) = \pm\sqrt{\frac{1-\cos\theta}{2}}, \qquad \cos(\theta/2) = \pm\sqrt{\frac{1+\cos\theta}{2}}.$$

Choose the sign by the quadrant containing $\theta/2$.

Exercises

1. Derive the formula for $\sin^2(\theta/2)$ from a double-angle formula.
2. Derive the formula for $\cos^2(\theta/2)$ from a double-angle formula.
3. Find $\sin(\pi/8)$ exactly.
4. Find $\cos(\pi/8)$ exactly.
5. Find $\sin(3\pi/8)$ exactly.
6. Find $\cos(3\pi/8)$ exactly.

7. If $\cos\theta = -3/5$ and $\theta/2$ is in the first quadrant, find $\sin(\theta/2)$.

8. If $\cos\theta = -3/5$ and $\theta/2$ is in the first quadrant, find $\cos(\theta/2)$.

9. Simplify $\sqrt{(1-\cos\theta)/2}$ when $\theta/2$ is in the second quadrant.

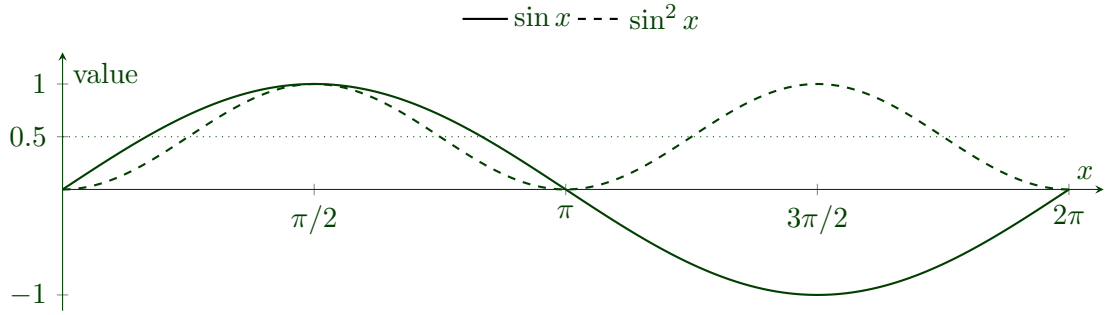
10. Find all signs required in the half-angle formulas when $\theta/2$ is in the third quadrant.

Squares of Sine and Cosine

Replacing θ by 2θ in the squared half-angle identities gives

$$\boxed{\sin^2\theta = \tfrac{1}{2}(1-\cos(2\theta))}, \qquad \boxed{\cos^2\theta = \tfrac{1}{2}(1+\cos(2\theta))}.$$

Here $\sin^2\theta$ means $(\sin\theta)^2$, not $\sin(\theta^2)$. The graph shows that squaring makes both the positive and negative humps of sine nonnegative, producing two humps per sine period.



For $\omega \neq 0$, $\sin^2(\omega t) = \frac{1}{2} - \frac{1}{2}\cos(2\omega t)$ has period $T = \pi/|\omega|$, half the period of $\sin(\omega t)$. Its oscillating part therefore has twice the frequency; it also has a constant (DC) component of $1/2$. Its average over any whole period is

$$\frac{1}{T}\int_{t_0}^{t_0+T} \sin^2(\omega t)\,dt = \frac{1}{2},$$

because the cosine integrates to zero over that period. The same conclusions hold for $\cos^2(\omega t)$. The restriction $\omega \neq 0$ matters: when $\omega = 0$, the squared sine is identically zero.

Additional Exercises: Squares of Sine and Cosine

11. Express $\sin^2(3t)$ and $\cos^2(3t)$ using cosines of double angles.

12. Find the mean value, fundamental period, and oscillation frequency in hertz of $\sin^2(5t)$.

13. Evaluate $\int_0^{2\pi} \sin^2 x\,dx$ using a power-reduction identity.

14. Find an antiderivative of $\cos^2(2x)$.

15. Simplify $3\sin^2\theta + 2\cos^2\theta$ to a constant plus one cosine, and find its average over $0 \le \theta \le 2\pi$.

8 Products of Trigonometric Functions

The addition formulas turn certain products into sums. The following three lemmas make the steps explicit.

Lemma 1. For every real u, $\sin^2 u + \cos^2 u = 1$.

Lemma 2. For any x, y, $(x+y)(x-y) = x^2 - y^2$.

Lemma 3. The half-angle formulas, with their arguments doubled, give

$$\cos^2 A = \tfrac{1}{2}\big(1 + \cos(2A)\big),$$
$$\sin^2 B = \tfrac{1}{2}\big(1 - \cos(2B)\big).$$

Now use the cosine addition formulas and then the lemmas:

$$\begin{aligned}
\cos(A-B)\cos(A+B) &= (\cos A\cos B + \sin A\sin B)(\cos A\cos B - \sin A\sin B) \\
&= (\cos A\cos B)^2 - (\sin A\sin B)^2 && \text{(Lemma 2)} \\
&= \cos^2 A(1-\sin^2 B) - (1-\cos^2 A)\sin^2 B && \text{(Lemma 1)} \\
&= \cos^2 A - \cos^2 A\sin^2 B - \sin^2 B + \cos^2 A\sin^2 B \\
&= \cos^2 A - \sin^2 B \\
&= \tfrac{1}{2}\big(1+\cos(2A)\big) - \tfrac{1}{2}\big(1-\cos(2B)\big) && \text{(Lemma 3)} \\
&= \tfrac{1}{2}\big(\cos(2A) + \cos(2B)\big).
\end{aligned}$$

Similarly, the sine addition formulas give

$$\begin{aligned}
\sin(A-B)\sin(A+B) &= (\sin A\cos B)^2 - (\cos A\sin B)^2 \\
&= \sin^2 A - \sin^2 B = \frac{\cos(2B) - \cos(2A)}{2}.
\end{aligned}$$

In particular, setting $A = (x+y)/2$ and $B = (y-x)/2$ yields the *product-to-sum* identities

$$\cos x\cos y = \tfrac{1}{2}\big[\cos(x-y) + \cos(x+y)\big],$$
$$\sin x\sin y = \tfrac{1}{2}\big[\cos(x-y) - \cos(x+y)\big].$$

To turn a sum into a product, add $\cos(A+B)$ and $\cos(A-B)$: $\cos(A+B) + \cos(A-B) = 2\cos A\cos B$. Taking $A = (\omega_1+\omega_2)t/2$ and $B = (\omega_1-\omega_2)t/2$ gives

$$\cos(\omega_1 t) + \cos(\omega_2 t) = 2\cos\left(\frac{\omega_1+\omega_2}{2}t\right)\cos\left(\frac{\omega_1-\omega_2}{2}t\right).$$

For example, $\cos(3t) + \cos(5t) = 2\cos(4t)\cos t$.

Exercises

1. Derive $\cos(A-B)\cos(A+B) = \cos^2 A - \sin^2 B$.
2. Derive $\sin(A-B)\sin(A+B) = \sin^2 A - \sin^2 B$.
3. Express $\cos x\cos y$ as a sum of cosines.
4. Express $\sin x\sin y$ as a difference of cosines.
5. Evaluate $\cos(\pi/12)\cos(5\pi/12)$ exactly.

6. Evaluate $\sin(\pi/12)\sin(5\pi/12)$ exactly.

7. Rewrite $\cos(3t) + \cos(5t)$ as a product.

8. Rewrite $\cos(2t) + \cos(6t)$ as a product.

9. Rewrite $\cos(7t)\cos(3t)$ as a sum.

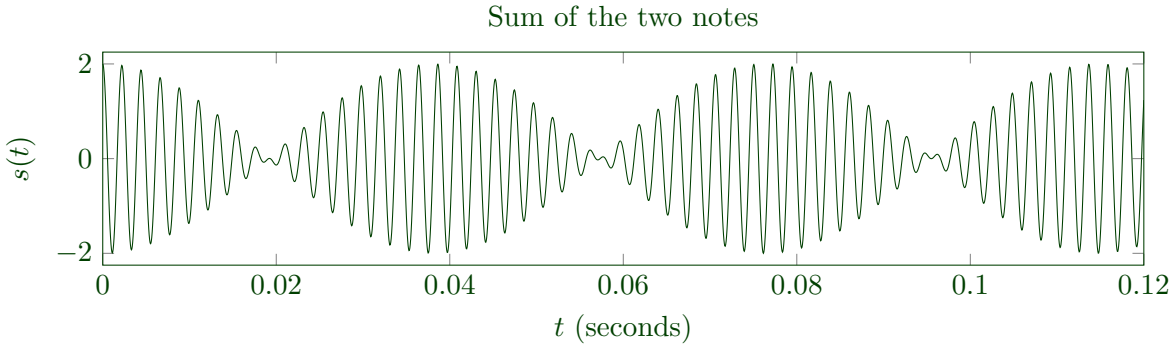
10. Rewrite $\sin(7t)\sin(3t)$ as a difference.

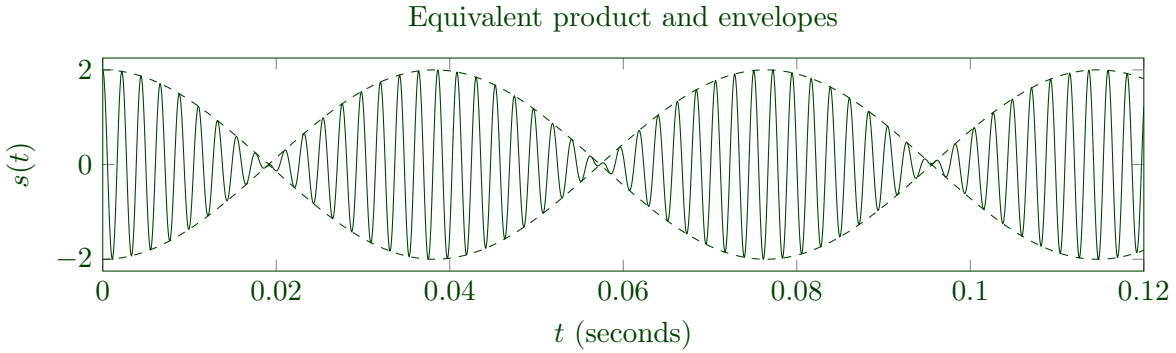
Adjacent Musical Notes and Beats

Take middle A to mean A4, the A above middle C. With the usual equal-tempered tuning, choose $f_1 = 440$ Hz and $f_2 = 440\,2^{1/12} \approx 466.164$ Hz for A4 and A♯4, respectively, and set $\omega_i = 2\pi f_i$. Then, with t in seconds,

$$\begin{aligned} s(t) &= \cos(\omega_1 t) + \cos(\omega_2 t) \\ &= 2\cos\big(\pi(f_2 - f_1)t\big)\cos\big(2\pi(f_1 + f_2)t/2\big) \\ &\approx 2\cos(2\pi \cdot 13.082t)\cos(2\pi \cdot 453.082t). \end{aligned}$$

The rapidly oscillating factor is multiplied by a slowly varying factor. The two plots below show the same waveform written as a sum and as a product; the dashed curves in the second plot are its upper and lower envelopes.





The amplitude envelope is $2|\cos(\pi(f_2 - f_1)t)|$, so amplitude maxima recur at the difference frequency $f_2 - f_1 \approx 26.164$ Hz. This is the beat rate, not the 13.082 Hz frequency of the signed cosine factor. Two side-by-side notes such as A and A♯, played together, commonly give a discordant

or rough sound. At this separation the fluctuations are rapid; perceived roughness also depends on timbre and listening conditions. For the note-frequency convention and sound demonstrations, see UNSW's note-frequency table and its beats demonstration.

9 The Substitution Formula for an Integral

For an indefinite integral, set $u = g(x)$ and $du = g'(x)\,dx$. Then

$$\int f(g(x))g'(x)\,dx = \int f(u)\,du.$$

For a definite integral, if $u = g(x)$, transform both the integrand and the limits:

$$\int_{x=a}^{x=b} f(g(x))g'(x)\,dx = \int_{u=g(a)}^{u=g(b)} f(u)\,du.$$

Here are three examples. First, with $u = 3t - 1$ and $du = 3\,dt$,

$$\int \cos(3t-1)\,dt = \frac{1}{3}\int \cos u\,du = \frac{1}{3}\sin(3t-1) + C.$$

Second, with $u = 2t + p$ and $du = 2\,dt$,

$$\int \sin(2t+p)\,dt = \frac{1}{2}\int \sin u\,du = -\frac{1}{2}\cos(2t+p) + C.$$

Finally, if $\omega \neq 0$, set $u = \mathrm{j}\omega t$, so $du = \mathrm{j}\omega\,dt$:

$$\int \mathrm{e}^{\mathrm{j}\omega t}\,dt = \frac{1}{\mathrm{j}\omega}\int \mathrm{e}^{u}\,du = \frac{\mathrm{e}^{\mathrm{j}\omega t}}{\mathrm{j}\omega} + C.$$

Exercises

1. Evaluate $\int 2x\cos(x^2)\,dx$ by substitution.
2. Evaluate $\int 3x^2\mathrm{e}^{x^3}\,dx$ by substitution.
3. Evaluate $\int \sin(5x)\,dx$ by substitution.
4. Evaluate $\int \cos(4t-1)\,dt$ by substitution.
5. Evaluate $\int \mathrm{e}^{2x+3}\,dx$ by substitution.
6. Evaluate $\int_0^1 2x\mathrm{e}^{x^2}\,dx$ by substitution.
7. Evaluate $\int_0^{\pi/2} \sin(2x)\,dx$ by substitution.
8. State a useful substitution for $\int x(x^2+1)^7\,dx$.
9. State a useful substitution for $\int \cos(3t+2)\,dt$.
10. Evaluate $\int (6x-2)\sin(3x^2-2x)\,dx$ by substitution.

10 Integrals of Sine, Cosine, and Exponentials

The basic antiderivatives are

$$\int \sin x\,dx = -\cos x + C, \qquad \int \cos x\,dx = \sin x + C, \qquad \int \mathrm{e}^x\,dx = \mathrm{e}^x + C.$$

Substitution gives the scaled forms

$$\int \sin(at+p)\,dt = -\frac{1}{a}\cos(at+p) + C,$$

$$\int \cos(at+p)\,dt = \frac{1}{a}\sin(at+p) + C, \qquad \int \mathrm{e}^{at+p}\,dt = \frac{1}{a}\mathrm{e}^{at+p} + C,$$

provided $a \neq 0$.

As an example involving a product, use the formula from Section 8:

$$\sin(mt)\sin(nt) = \tfrac{1}{2}\big[\cos((m-n)t) - \cos((m+n)t)\big].$$

Let m, n be nonzero integers, and write $I_{m,n} = \int_0^{2\pi} \sin(mt)\sin(nt)\,dt$. If $m \neq n$ and $m \neq -n$, direct integration gives

$$\begin{aligned} I_{m,n} &= \frac{1}{2}\int_0^{2\pi} \big[\cos((m-n)t) - \cos((m+n)t)\big]\,dt \\ &= \frac{1}{2}\left[\frac{\sin(2\pi(m-n))}{m-n} - \frac{\sin(2\pi(m+n))}{m+n}\right] = 0, \end{aligned}$$

because both sine numerators vanish. When $m = n$, the first cosine is 1, so

$$I_{m,m} = \frac{1}{2}\int_0^{2\pi} [1 - \cos(2mt)]\,dt = \pi.$$

When $m = -n$,

$$\sin(nt) = \sin(-mt) = -\sin(mt).$$

Lemma 3 from Section 8 gives

$$\sin^2(mt) = \tfrac{1}{2}[1 - \cos(2mt)],$$

so

$$\begin{aligned} I_{m,-m} &= \int_0^{2\pi} \sin(mt)\sin(-mt)\,dt \\ &= -\int_0^{2\pi} \sin^2(mt)\,dt \\ &= -\frac{1}{2}\int_0^{2\pi} [1 - \cos(2mt)]\,dt \quad \text{(Lemma 3)} \\ &= -\frac{1}{2}\left[t - \frac{\sin(2mt)}{2m}\right]_0^{2\pi} \\ &= -\pi + \frac{\sin(4\pi m)}{4m} = -\pi, \end{aligned}$$

because m is an integer. If either index is zero, the integral is zero. Thus, for all integers m, n,

$$I_{m,n} = \begin{cases} \pi, & m = n \neq 0, \\ -\pi, & m = -n \neq 0, \\ 0, & \text{otherwise.} \end{cases}$$

For positive integers, only the first and third cases can occur.

Exercises

1. Evaluate $\int \sin x \, dx$.
2. Evaluate $\int \cos x \, dx$.
3. Evaluate $\int \mathrm{e}^x \, dx$.
4. Evaluate $\int \sin(3t) \, dt$.
5. Evaluate $\int \cos(2x - 1) \, dx$.
6. Evaluate $\int \mathrm{e}^{5x} \, dx$.
7. Evaluate $\int \sin(\omega t + p) \, dt$, where $\omega \neq 0$.
8. Evaluate $\int_0^{\pi} \sin x \, dx$.
9. Evaluate $\int_0^{\pi/2} \cos(2x) \, dx$.
10. Find an antiderivative of $3\mathrm{e}^{2t} - 4\sin t + 5\cos(3t)$.

11 Integration by Parts

The product rule $(uv)' = u'v + uv'$ gives, after integration and rearrangement,

$$\boxed{\int u \, dv = uv - \int v \, du}.$$

Choose u so that differentiating it simplifies the remaining integral, and choose dv so that it is easy to integrate. For definite integrals,

$$\int_a^b u(x)v'(x) \, dx = [u(x)v(x)]_a^b - \int_a^b v(x)u'(x) \, dx.$$

Worked Examples

For $\int x \sin x \, dx$, take $u = x$, $dv = \sin x \, dx$, so $du = dx$ and $v = -\cos x$. Then

$$\int x \sin x \, dx = -x\cos x + \int \cos x \, dx = -x\cos x + \sin x + C.$$

For $\int x^2 \sin x \, dx$, repeat the process:

$$\begin{aligned} \int x^2 \sin x \, dx &= -x^2 \cos x + 2\int x \cos x \, dx \\ &= -x^2 \cos x + 2x \sin x + 2\cos x + C. \end{aligned}$$

Likewise, $u = x$, $dv = \mathrm{e}^x dx$ gives

$$\int x\mathrm{e}^x \, dx = x\mathrm{e}^x - \int \mathrm{e}^x \, dx = (x-1)\mathrm{e}^x + C.$$

For the cyclic example $I = \int e^x \sin x\, dx$, two integrations by parts return the original integral. Suppressing constants during the calculation,

$$I = e^x \sin x - \int e^x \cos x\, dx = e^x \sin x - e^x \cos x - I.$$

Thus $2I = e^x(\sin x - \cos x)$, and

$$\int e^x \sin x\, dx = \tfrac{1}{2} e^x(\sin x - \cos x) + C.$$

Differentiate each answer to check it.

Exercises

1. Evaluate $\int x \sin x\, dx$ by parts.
2. Evaluate $\int x^2 \sin x\, dx$ by parts twice.
3. Evaluate $\int \sin x\, e^x\, dx$ using two integrations by parts.
4. Evaluate $\int x e^x\, dx$.
5. Evaluate $\int_0^{\pi} x \sin x\, dx$.
6. Evaluate $\int x \cos(2x)\, dx$.

12 Constant-Coefficient First-Order ODEs

For the homogeneous equation

$$y' + ay = 0,$$

try $y = e^{rt}$. Substitution gives $(r + a)e^{rt} = 0$, hence $r = -a$ and

$$y = Ce^{-at}.$$

The polynomial $p(r) = r + a$ is called the *characteristic polynomial* of this equation. More generally, substitution of $y = e^{rt}$ into a constant-coefficient homogeneous ODE

$$c_n y^{(n)} + c_{n-1} y^{(n-1)} + \cdots + c_1 y' + c_0 y = 0$$

gives

$$\big(c_n r^n + c_{n-1} r^{n-1} + \cdots + c_1 r + c_0\big) e^{rt} = 0.$$

Thus the corresponding characteristic polynomial is

$$p(r) = c_n r^n + c_{n-1} r^{n-1} + \cdots + c_1 r + c_0,$$

and its roots identify the exponential trial solutions. For $y' + ay = f(t)$, a solution has the form $y = y_h + y_p$, where y_h solves the homogeneous equation and y_p is any particular solution of the full equation. An initial condition determines the remaining constant.

Exercises

1. Solve $y' + 2y = 0$.
2. Solve $y' - 3y = 0$.
3. Solve $y' + 4y = 0$ subject to $y(0) = 5$.
4. Solve $y' - 2y = 0$ subject to $y(0) = -1$.
5. Verify that $y = Ce^{-at}$ solves $y' + ay = 0$.
6. Find the characteristic equation for $y' + 7y = 0$.
7. Find a constant particular solution of $y' + 2y = 6$.
8. Solve $y' + 2y = 6$.
9. Solve $y' + 2y = 6$ subject to $y(0) = 1$.
10. Find a particular solution of $y' - 3y = 6e^{2t}$.

13 Constant-Coefficient Second-Order ODEs

For

$$y'' + ay' + by = 0,$$

try $y = e^{rt}$. This produces the characteristic equation

$$r^2 + ar + b = 0.$$

If its roots are distinct real numbers r_1, r_2, then $y = C_1e^{r_1t} + C_2e^{r_2t}$. For a repeated root r, use $y = (C_1 + C_2t)e^{rt}$. For complex roots $\alpha \pm j\beta$, use

$$y = e^{\alpha t}(C_1 \cos \beta t + C_2 \sin \beta t).$$

For $y'' + ay' + by = f(t)$, add a particular solution to the homogeneous solution, then use two initial conditions to determine C_1, C_2.

Worked examples

For $y'' + \omega^2 y = 0$ with real $\omega \neq 0$, the characteristic equation is $r^2 + \omega^2 = 0$. Its roots are $r = \pm j\omega$, so the general real solution is

$$y(t) = C_1 \cos(\omega t) + C_2 \sin(\omega t).$$

If $\omega = 0$, the equation is $y'' = 0$ and its solution is $y(t) = C_1 + C_2t$.

For $y'' + y' + 2y = 0$, the characteristic equation is $r^2 + r + 2 = 0$. The quadratic formula gives $r = (-1 \pm j\sqrt{7})/2$, so

$$y(t) = e^{-t/2}\left(C_1 \cos \frac{\sqrt{7}t}{2} + C_2 \sin \frac{\sqrt{7}t}{2}\right).$$

For $y'' + y' - 2y = 0$, the characteristic polynomial factors as $r^2 + r - 2 = (r - 1)(r + 2)$. The distinct real roots are 1 and -2, giving

$$y(t) = C_1e^t + C_2e^{-2t}.$$

Exercises

1. Find the characteristic equation for $y'' + 5y' + 6y = 0$.
2. Solve $y'' + 5y' + 6y = 0$.
3. Solve $y'' - 4y' + 4y = 0$.
4. Solve $y'' + 4y = 0$.
5. Solve $y'' + 2y' + 5y = 0$.
6. Solve $y'' + 5y' + 6y = 0$ subject to $y(0) = 1$ and $y'(0) = 0$.
7. Find the general solution when the characteristic roots are $-2 \pm 3\mathrm{j}$.
8. Solve $y'' + 3y' + 2y = 0$ subject to $y(0) = 1$ and $y'(0) = -1$.
9. Solve $y'' - y = 0$ subject to $y(0) = 0$ and $y'(0) = 2$.
10. Solve $y'' + 4y = 8$ subject to $y(0) = 0$ and $y'(0) = 0$.

14 Inhomogeneous First-Order Constant-Coefficient ODEs

After dividing by a nonzero leading coefficient, write the equation as

$$y' + ay = f(t),$$

where a is constant. Here $f(t)$ is either $\sin(\omega t)$ or a polynomial of degree at most two. The general solution is

$$\boxed{y(t) = y_p(t) + y_h(t) = y_p(t) + C\mathrm{e}^{-at}},$$

where y_p is one solution of the inhomogeneous equation and y_h is the family of all solutions of the homogeneous equation. Indeed, linearity shows that their sum solves the full equation, and subtracting y_p from any full solution gives a homogeneous solution. An initial condition fixes C.

Polynomial Forcing

For $a \neq 0$ and $f(t) = q_2t^2 + q_1t + q_0$, choose a quadratic trial and differentiate:

$$y_p = At^2 + Bt + D, \qquad y_p' = 2At + B.$$

Substitute these expressions into $y_p' + ay_p = f(t)$:

$$(2At + B) + a(At^2 + Bt + D) = q_2t^2 + q_1t + q_0.$$

Expand and group by powers of t:

$$\begin{aligned} 2At + B + aAt^2 + aBt + aD &= q_2t^2 + q_1t + q_0, \\ (aA)t^2 + (2A + aB)t + (B + aD) &= q_2t^2 + q_1t + q_0. \end{aligned}$$

Equality for every t requires matching coefficients:

$$aA = q_2, \qquad 2A + aB = q_1, \qquad B + aD = q_0.$$

Solve in order:

$$A = \frac{q_2}{a}, \qquad B = \frac{q_1 - 2A}{a}, \qquad D = \frac{q_0 - B}{a}.$$

This determines $y_p(t) = At^2 + Bt + D$. Adding the homogeneous solution gives the general solution

$$\boxed{y(t) = At^2 + Bt + D + C\mathrm{e}^{-at}},$$

with A, B, D as above and C arbitrary.

For example, in $y' + 2y = t^2$, substitute the same trial:

$$(2At + B) + 2(At^2 + Bt + D) = t^2,$$
$$2At^2 + (2A + 2B)t + (B + 2D) = t^2 + 0t + 0.$$

Thus

$$\begin{aligned} 2A = 1 \quad &\Longrightarrow \quad A = \tfrac{1}{2}, \\ 2A + 2B = 0 \quad &\Longrightarrow \quad 1 + 2B = 0 \quad \Longrightarrow \quad B = -\tfrac{1}{2}, \\ B + 2D = 0 \quad &\Longrightarrow \quad -\tfrac{1}{2} + 2D = 0 \quad \Longrightarrow \quad D = \tfrac{1}{4}. \end{aligned}$$

Consequently,

$$y_p = \tfrac{1}{2}t^2 - \tfrac{1}{2}t + \tfrac{1}{4}, \qquad y = y_p + C\mathrm{e}^{-2t}.$$

If $a = 0$, integrate $y' = q_2t^2 + q_1t + q_0$ directly:

$$y_p(t) = \frac{q_2}{3}t^3 + \frac{q_1}{2}t^2 + q_0t.$$

The homogeneous solutions are constants, so the general solution is

$$y(t) = \frac{q_2}{3}t^3 + \frac{q_1}{2}t^2 + q_0t + C.$$

Sinusoidal Forcing

Suppose you want to find the general solution to the equation

$$y'(t) + a\,y(t) = \sin(\omega t).$$

Here a and ω are real constants, with $\omega \neq 0$. Guess a particular solution $y_p(t) = A\sin(\omega t) + B\cos(\omega t)$. Both sine and cosine are needed because differentiation interchanges them. Calculate the derivative:

$$y_p = A\sin(\omega t) + B\cos(\omega t),$$
$$y_p' = A\omega\cos(\omega t) - B\omega\sin(\omega t).$$

Substitute into $y_p' + ay_p = \sin(\omega t)$:

$$A\omega\cos(\omega t) - B\omega\sin(\omega t) + a\big[A\sin(\omega t) + B\cos(\omega t)\big] = \sin(\omega t).$$

Group the sine terms and cosine terms:

$$(aA - \omega B)\sin(\omega t) + (\omega A + aB)\cos(\omega t) = \sin(\omega t).$$

Matching their coefficients gives

$$aA - \omega B = 1, \qquad \omega A + aB = 0.$$

Multiply the first equation by a, the second by ω, and add:

$$(a^2 A - a\omega B) + (\omega^2 A + a\omega B) = a, \qquad (a^2 + \omega^2)A = a.$$

Multiply the first equation by $-\omega$, the second by a, and add:

$$(-a\omega A + \omega^2 B) + (a\omega A + a^2 B) = -\omega, \qquad (a^2 + \omega^2)B = -\omega.$$

Therefore

$$A = \frac{a}{a^2 + \omega^2}, \qquad B = -\frac{\omega}{a^2 + \omega^2}, \qquad y_p = \frac{a\sin(\omega t) - \omega\cos(\omega t)}{a^2 + \omega^2}.$$

The homogeneous equation $y' + ay = 0$ has the family $y_h = Ce^{-at}$. Therefore the general solution of the original equation is

$$\boxed{y(t) = \frac{a\sin(\omega t) - \omega\cos(\omega t)}{a^2 + \omega^2} + Ce^{-at}}.$$

For example, for $y' + 2y = \sin(3t)$, take $y_p = A\sin(3t) + B\cos(3t)$, so $y_p' = 3A\cos(3t) - 3B\sin(3t)$. Substitution and grouping give

$$3A\cos(3t) - 3B\sin(3t) + 2[A\sin(3t) + B\cos(3t)] = \sin(3t),$$
$$(2A - 3B)\sin(3t) + (3A + 2B)\cos(3t) = \sin(3t).$$

Thus $2A - 3B = 1$ and $3A + 2B = 0$. Twice the first equation plus three times the second gives $13A = 2$; minus three times the first plus twice the second gives $13B = -3$. Hence

$$y = \frac{2\sin(3t) - 3\cos(3t)}{13} + Ce^{-2t}.$$

If $\omega = 0$, the forcing is zero and the general solution is $y(t) = Ce^{-at}$.

Exercises

Find the general solution unless initial data are supplied.

1. $y' + 2y = 1$.
2. $y' - y = t$.
3. $y' + 2y = t^2$.
4. $y' = t^2 + 2t + 1$, with $y(0) = 2$.
5. $y' + y = \sin(2t)$.
6. $y' + 2y = \sin t$, with $y(0) = 0$.

15 Inhomogeneous Second-Order Constant-Coefficient ODEs

After division by the nonzero coefficient of y'', consider

$$y'' + ay' + by = f(t),$$

where a, b are constants and $f(t)$ is either $\sin(\omega t)$ or a polynomial of degree at most two. As in the first-order case,

$$\boxed{y(t) = y_p(t) + y_h(t)}.$$

Here y_p is one particular solution of the full equation and y_h is the entire two-constant homogeneous family found from $r^2 + ar + b = 0$. Linearity proves both that every such sum is a solution and that every solution has this form. Two initial conditions determine the two constants.

Polynomial Forcing and Repeated Trials

If $b \neq 0$ and $f(t) = q_2t^2 + q_1t + q_0$, choose a quadratic trial and calculate both derivatives:

$$y_p = At^2 + Bt + D, \qquad y_p' = 2At + B, \qquad y_p'' = 2A.$$

Substitute into $y_p'' + ay_p' + by_p = f(t)$:

$$2A + a(2At + B) + b(At^2 + Bt + D) = q_2t^2 + q_1t + q_0.$$

Expand and group:

$$\begin{aligned} 2A + 2aAt + aB + bAt^2 + bBt + bD &= q_2t^2 + q_1t + q_0, \\ (bA)t^2 + (2aA + bB)t + (2A + aB + bD) &= q_2t^2 + q_1t + q_0. \end{aligned}$$

Matching coefficients and solving in order gives

$$\begin{aligned} bA = q_2 \quad &\Longrightarrow \quad A = \frac{q_2}{b}, \\ 2aA + bB = q_1 \quad &\Longrightarrow \quad B = \frac{q_1 - 2aA}{b}, \\ 2A + aB + bD = q_0 \quad &\Longrightarrow \quad D = \frac{q_0 - 2A - aB}{b}. \end{aligned}$$

Thus $y_p(t) = At^2 + Bt + D$ with these coefficients. The general solution is

$$\boxed{y(t) = At^2 + Bt + D + y_h(t)},$$

where y_h is the full two-constant family obtained from $r^2 + ar + b = 0$:

$$y_h(t) = \begin{cases} C_1\mathrm{e}^{r_1t} + C_2\mathrm{e}^{r_2t}, & r_1, r_2 \text{ distinct real roots}, \\ (C_1 + C_2t)\mathrm{e}^{rt}, & r \text{ a repeated real root}, \\ \mathrm{e}^{\alpha t}(C_1\cos\beta t + C_2\sin\beta t), & \alpha \pm \mathrm{j}\beta \text{ complex roots}, \ \beta > 0. \end{cases}$$

For example, for $y'' + 3y' + 2y = t^2$, substitution and grouping give

$$\begin{aligned} 2A + 3(2At + B) + 2(At^2 + Bt + D) &= t^2, \\ 2At^2 + (6A + 2B)t + (2A + 3B + 2D) &= t^2 + 0t + 0. \end{aligned}$$

Thus

$$\begin{aligned} 2A = 1 &\implies A = \tfrac{1}{2}, \\ 6A + 2B = 0 &\implies 3 + 2B = 0 \implies B = -\tfrac{3}{2}, \\ 2A + 3B + 2D = 0 &\implies 1 - \tfrac{9}{2} + 2D = 0 \implies D = \tfrac{7}{4}. \end{aligned}$$

Consequently,

$$y_p = \tfrac{1}{2}t^2 - \tfrac{3}{2}t + \tfrac{7}{4}, \qquad y = y_p + C_1 \mathrm{e}^{-t} + C_2 \mathrm{e}^{-2t}.$$

If $b = 0$, a constant already solves the homogeneous equation. Multiply the polynomial trial by t when $a \neq 0$, or by t^2 when $a = 0$. More generally, multiply by t^m, where m is the multiplicity of the characteristic root zero. For instance, for $y'' = t^2$, try

$$y_p = t^2(At^2 + Bt + D) = At^4 + Bt^3 + Dt^2.$$

Then

$$y_p' = 4At^3 + 3Bt^2 + 2Dt, \qquad y_p'' = 12At^2 + 6Bt + 2D.$$

Substitution gives $12At^2 + 6Bt + 2D = t^2 + 0t + 0$, so $A = 1/12$, $B = 0$, $D = 0$. Hence $y = t^4/12 + C_1 + C_2 t$.

Sinusoidal Forcing

Suppose you want to find the general solution to the equation

$$y''(t) + a\, y'(t) + b\, y(t) = \sin(\omega t).$$

Here a, b, ω are real constants, with $\omega \neq 0$. Guess a particular solution $y_p(t) = A\sin(\omega t) + B\cos(\omega t)$, and differentiate twice:

$$\begin{aligned} y_p &= A\sin(\omega t) + B\cos(\omega t), \\ y_p' &= A\omega\cos(\omega t) - B\omega\sin(\omega t), \\ y_p'' &= -A\omega^2\sin(\omega t) - B\omega^2\cos(\omega t). \end{aligned}$$

Substitute into $y_p'' + ay_p' + by_p = \sin(\omega t)$:

$$\begin{aligned} &- A\omega^2\sin(\omega t) - B\omega^2\cos(\omega t) \\ &\quad + a[A\omega\cos(\omega t) - B\omega\sin(\omega t)] \\ &\quad + b[A\sin(\omega t) + B\cos(\omega t)] = \sin(\omega t). \end{aligned}$$

Grouping the sine and cosine terms gives

$$\big[(b - \omega^2)A - a\omega B\big]\sin(\omega t) + \big[a\omega A + (b - \omega^2)B\big]\cos(\omega t) = \sin(\omega t).$$

Write $d = b - \omega^2$. Matching coefficients yields

$$dA - a\omega B = 1, \qquad a\omega A + dB = 0.$$

Multiply the first equation by d, the second by $a\omega$, and add:

$$(d^2A - da\omega B) + (a^2\omega^2 A + da\omega B) = d, \qquad (d^2 + a^2\omega^2)A = d.$$

Multiply the first equation by $-a\omega$, the second by d, and add:

$$(-da\omega A + a^2\omega^2 B) + (da\omega A + d^2 B) = -a\omega, \qquad (d^2 + a^2\omega^2)B = -a\omega.$$

If $d^2 + a^2\omega^2 \neq 0$, divide to obtain

$$A = \frac{d}{d^2 + a^2\omega^2}, \qquad B = -\frac{a\omega}{d^2 + a^2\omega^2}, \qquad y_p = \frac{d\sin(\omega t) - a\omega\cos(\omega t)}{d^2 + a^2\omega^2}.$$

Add the homogeneous family to obtain the general solution:

$$\boxed{y(t) = \frac{d\sin(\omega t) - a\omega\cos(\omega t)}{d^2 + a^2\omega^2} + y_h(t)}, \qquad d = b - \omega^2,$$

where $y_h(t)$ is the two-constant family displayed above, determined by the roots of $r^2 + ar + b = 0$.

For example, for $y'' + 3y' + 2y = \sin t$, use

$$y_p = A\sin t + B\cos t, \quad y_p' = A\cos t - B\sin t, \quad y_p'' = -A\sin t - B\cos t.$$

Substitution and grouping give

$$(-A\sin t - B\cos t) + 3(A\cos t - B\sin t) + 2(A\sin t + B\cos t) = \sin t,$$
$$(A - 3B)\sin t + (3A + B)\cos t = \sin t.$$

Thus $A - 3B = 1$ and $3A + B = 0$. From the second equation, $B = -3A$; the first then gives $A + 9A = 1$, so $A = 1/10$ and $B = -3/10$. Therefore

$$y = \frac{\sin t - 3\cos t}{10} + C_1 e^{-t} + C_2 e^{-2t}.$$

For real coefficients and $\omega \neq 0$, the denominator vanishes precisely when $a = 0$ and $b = \omega^2$. The usual trial then fails; the next section derives the general solution for this resonant case. If $\omega = 0$, the forcing is zero and the general solution is simply $y(t) = y_h(t)$.

Exercises

Find the general solution unless initial data are supplied.

1. $y'' + 3y' + 2y = 1$.
2. $y'' + 3y' + 2y = t$.
3. $y'' + 3y' + 2y = t^2$.
4. $y'' = t^2$, with $y(0) = 1$ and $y'(0) = 2$.
5. $y'' + y' = t$.
6. $y'' + 4y = \sin t$.
7. $y'' + y = \sin t$.
8. $y'' + 3y' + 2y = \sin t$, with $y(0) = y'(0) = 0$.

16 Resonance

Suppose you want to find the general solution to

$$y''(t) + \omega_0^2 y(t) = \sin(\omega_0 t), \qquad \omega_0 > 0.$$

The associated homogeneous equation has general solution

$$y_h(t) = C_1 \cos(\omega_0 t) + C_2 \sin(\omega_0 t).$$

The forcing oscillates at the same angular frequency ω_0 as these homogeneous oscillations. This is resonance for an undamped oscillator. If we guess $y_p = A\sin(\omega_0 t) + B\cos(\omega_0 t)$, then $y_p'' + \omega_0^2 y_p = 0$, regardless of A, B, so this guess cannot produce the required nonzero right-hand side.

Deriving a Resonant Particular and General Solution

Multiply the trial by t:

$$\begin{aligned} y_p &= t[A\sin(\omega_0 t) + B\cos(\omega_0 t)], \\ y_p' &= A\sin(\omega_0 t) + B\cos(\omega_0 t) + t[A\omega_0\cos(\omega_0 t) - B\omega_0\sin(\omega_0 t)], \\ y_p'' &= 2A\omega_0\cos(\omega_0 t) - 2B\omega_0\sin(\omega_0 t) - \omega_0^2 t[A\sin(\omega_0 t) + B\cos(\omega_0 t)]. \end{aligned}$$

Substitute into $y_p'' + \omega_0^2 y_p = \sin(\omega_0 t)$:

$$\begin{aligned} &2A\omega_0\cos(\omega_0 t) - 2B\omega_0\sin(\omega_0 t) \\ &\quad - \omega_0^2 t[A\sin(\omega_0 t) + B\cos(\omega_0 t)] + \omega_0^2 t[A\sin(\omega_0 t) + B\cos(\omega_0 t)] = \sin(\omega_0 t). \end{aligned}$$

The terms containing t cancel, leaving

$$(-2B\omega_0)\sin(\omega_0 t) + (2A\omega_0)\cos(\omega_0 t) = \sin(\omega_0 t).$$

Hence $-2B\omega_0 = 1$, $2A\omega_0 = 0$, so $B = -1/(2\omega_0)$ and $A = 0$. Thus

$$y_p = -\frac{t\cos(\omega_0 t)}{2\omega_0}, \qquad y = y_p + C_1\cos(\omega_0 t) + C_2\sin(\omega_0 t).$$

The factor t in y_p makes its oscillation amplitude grow linearly with time. The homogeneous terms have bounded amplitude and cannot cancel this growth for all time. This conclusion concerns the undamped equation above.

Worked Example: Matching Frequencies

Find the general solution of $y'' + 4y = \sin(2t)$, and then impose $y(0) = y'(0) = 0$. The homogeneous angular frequency is 2, exactly the forcing angular frequency. With $\omega_0 = 2$, the derivation gives

$$y_p = -\frac{t}{4}\cos(2t), \qquad \boxed{y(t) = -\frac{t}{4}\cos(2t) + C_1\cos(2t) + C_2\sin(2t)}.$$

To verify the particular solution explicitly, differentiate:

$$\begin{aligned} y_p' &= -\tfrac{1}{4}\cos(2t) + \tfrac{t}{2}\sin(2t), \\ y_p'' &= \sin(2t) + t\cos(2t). \end{aligned}$$

Substitute and cancel the cosine terms:

$$y_p'' + 4y_p = \sin(2t) + t\cos(2t) - t\cos(2t) = \sin(2t).$$

For the initial conditions, $y(0) = C_1 = 0$ and $y'(0) = -1/4 + 2C_2 = 0$, so $C_2 = 1/8$. The solution of this initial-value problem is therefore

$$y(t) = -\frac{t}{4}\cos(2t) + \frac{1}{8}\sin(2t).$$

Exercises

1. Find the general solution of $y'' + y = \sin t$.
2. Find the general solution of $y'' + 9y = \sin(3t)$.
3. Solve $y'' + 4y = \sin(2t)$ with $y(0) = 1$ and $y'(0) = 0$.
4. For $y'' + 16y = \sin(\omega t)$ with $\omega > 0$, find the resonant value of ω and the general solution at that value.
5. For $y'' + 9y = \sin(2t)$ and $y'' + 9y = \sin(3t)$, identify which equation is resonant, find both general solutions, and explain why one has bounded solutions while the other has growing oscillations.

A Answers to the Exercises

The numbering restarts in each section, just as it does in the exercises. In antiderivatives, C denotes an arbitrary constant.

A.1 The Quadratic Formula

1. $x = 2, 3$.
2. $x = -2$ (a repeated root).
3. $x = -1 \pm 2\mathrm{j}$.
4. $x = 2, -\frac{1}{2}$.

A.2 Basic Complex Numbers

1. $a = 3$, $b = -7$; $\operatorname{Re} z = 3$ and $\operatorname{Im} z = -7$.
2. $7 - \mathrm{j}$.
3. $22 + 7\mathrm{j}$.
4. $2\mathrm{j}$.
5. $-4 - 9\mathrm{j}$.
6. 5.
7. $\frac{1}{2} + \frac{1}{2}\mathrm{j}$.

8. $-\frac{1}{5} + \frac{8}{5}\mathrm{j}$.
9. $(2-5\mathrm{j})(2+5\mathrm{j}) = 4+25 = 29 = |2-5\mathrm{j}|^2$.
10. Equating real and imaginary parts gives $a-b=3$ and $a+b=-1$; thus $a=1$, $b=-2$.
11. $2\mathrm{e}^{\mathrm{j}\pi/3}$.
12. $\sqrt{2}\,\mathrm{e}^{3\mathrm{j}\pi/4}$.

A.3 Arguments and Polar Multiplication

1. $\arg(-1-\mathrm{j}) = -3\pi/4$; all arguments are $-3\pi/4 + 2k\pi$, $k \in \mathbb{Z}$.
2. $\arg(-3) = \pi$ and $\arg(2\mathrm{j}) = \pi/2$. Zero has no direction, so no unique principal argument can be assigned by the polar direction rule.
3. $r=2$, $\theta = -\pi/6$, so $2\mathrm{e}^{-\mathrm{j}\pi/6}$.
4. $r = 3\sqrt{2}$, $\theta = 3\pi/4$, so $3\sqrt{2}\mathrm{e}^{3\mathrm{j}\pi/4}$.
5. $5(0+\mathrm{j}) = 5\mathrm{j}$.
6. $2(-1/2 - \mathrm{j}\sqrt{3}/2) = -1 - \sqrt{3}\mathrm{j}$.
7. $6\mathrm{e}^{5\mathrm{j}\pi/12}$; magnitude 6, principal argument $5\pi/12$.
8. $1+\mathrm{j} = \sqrt{2}\mathrm{e}^{\mathrm{j}\pi/4}$ and $1-\sqrt{3}\mathrm{j} = 2\mathrm{e}^{-\mathrm{j}\pi/3}$. The product is $2\sqrt{2}\mathrm{e}^{-\mathrm{j}\pi/12} = (1+\sqrt{3}) + (1-\sqrt{3})\mathrm{j}$.
9. $\mathrm{j}z = 3+2\mathrm{j}$ (counterclockwise $90°$); $z/\mathrm{j} = -3-2\mathrm{j}$ (clockwise $90°$). Both have magnitude $\sqrt{13}$.
10. $\sqrt{3}+\mathrm{j} = 2\mathrm{e}^{\mathrm{j}\pi/6}$, so its cube is $8\mathrm{e}^{\mathrm{j}\pi/2} = 8\mathrm{j}$.

A.4 Euler's Formula

1. j.
2. -1.
3. $\sqrt{2}\,\mathrm{e}^{\mathrm{j}\pi/4}$.
4. $2\mathrm{e}^{2\mathrm{j}\pi/3}$.
5. $2 - 2\sqrt{3}\,\mathrm{j}$.
6. $-\mathrm{j}$.
7. $\mathrm{e}^{\mathrm{j}(a+b)} = \cos(a+b) + \mathrm{j}\sin(a+b)$.
8. Magnitude 3; one argument is $\pi/6$.

A.5 Trigonometric Addition Formulas

1. $\sin(\pi/3 - \pi/4) = (\sqrt{6} - \sqrt{2})/4$.
2. $\cos(\pi/3 - \pi/4) = (\sqrt{6} + \sqrt{2})/4$.
3. $(\sqrt{3}/2)\sin x + (1/2)\cos x$.
4. $(1/2)\cos x + (\sqrt{3}/2)\sin x$.
5. $2\sin x \cos y$.
6. $2\cos x \cos y$.
7. Since $\cos(-b) = \cos b$ and $\sin(-b) = -\sin b$, $\sin(a - b) = \sin a \cos b - \cos a \sin b$.
8. $(\sqrt{6} + \sqrt{2})/4$.
9. $(\sqrt{2} - \sqrt{6})/4$.
10. $56/65$.

A.6 Double-Angle Formulas

1. Set $b = a$: $\sin(2a) = \sin a \cos a + \cos a \sin a = 2\sin a \cos a$.
2. Set $b = a$: $\cos(2a) = \cos^2 a - \sin^2 a$.
3. $1 - 2\sin^2 x$.
4. $2\cos^2 x - 1$.
5. $\sin(2x)$.
6. $\cos(2x)$.
7. $\sqrt{3}/2$.
8. $7/25$.
9. $120/169$.
10. $x = \pi/4$ or $x = 5\pi/4$.

A.7 Half-Angle Formulas

1. From $\cos\theta = 1 - 2\sin^2(\theta/2)$, $\sin^2(\theta/2) = (1 - \cos\theta)/2$.
2. From $\cos\theta = 2\cos^2(\theta/2) - 1$, $\cos^2(\theta/2) = (1 + \cos\theta)/2$.
3. $\sqrt{2 - \sqrt{2}}/2$.
4. $\sqrt{2 + \sqrt{2}}/2$.
5. $\sqrt{2 + \sqrt{2}}/2$.
6. $\sqrt{2 - \sqrt{2}}/2$.
7. $2/\sqrt{5}$.
8. $1/\sqrt{5}$.
9. $\sin(\theta/2)$, since sine is positive in the second quadrant.

10. Both signs are negative: $\sin(\theta/2) = -\sqrt{(1-\cos\theta)/2}$ and $\cos(\theta/2) = -\sqrt{(1+\cos\theta)/2}$.

Additional Exercises: Squares of Sine and Cosine

11. $\sin^2(3t) = \frac{1}{2}(1-\cos 6t)$; $\cos^2(3t) = \frac{1}{2}(1+\cos 6t)$.
12. $\sin^2(5t) = \frac{1}{2} - \frac{1}{2}\cos 10t$: mean $1/2$, period $\pi/5$, frequency $5/\pi$ Hz when t is in seconds.
13. $\int_0^{2\pi}(1-\cos 2x)/2\,dx = [x/2 - \sin(2x)/4]_0^{2\pi} = \pi$.
14. $\int(1+\cos 4x)/2\,dx = x/2 + \sin(4x)/8 + C$.
15. $3(1-\cos 2\theta)/2 + 2(1+\cos 2\theta)/2 = 5/2 - \cos(2\theta)/2$; the average is $5/2$.

A.8 Products of Trigonometric Functions

1. Multiply $(\cos A\cos B + \sin A\sin B)(\cos A\cos B - \sin A\sin B)$, then use $\cos^2 B = 1-\sin^2 B$ and $\sin^2 A = 1-\cos^2 A$. The result is $\cos^2 A - \sin^2 B$.
2. Multiply $(\sin A\cos B - \cos A\sin B)(\sin A\cos B + \cos A\sin B)$ and simplify to $\sin^2 A - \sin^2 B$.
3. $\frac{1}{2}[\cos(x-y) + \cos(x+y)]$.
4. $\frac{1}{2}[\cos(x-y) - \cos(x+y)]$.
5. $\cos^2(\pi/4) - \sin^2(\pi/6) = 1/4$.
6. $\sin^2(\pi/4) - \sin^2(\pi/6) = 1/4$.
7. $2\cos(4t)\cos t$.
8. $2\cos(4t)\cos(2t)$.
9. $\frac{1}{2}[\cos(4t) + \cos(10t)]$.
10. $\frac{1}{2}[\cos(4t) - \cos(10t)]$.

A.9 The Substitution Formula for an Integral

1. $\sin(x^2) + C$.
2. $\mathrm{e}^{x^3} + C$.
3. $-\frac{1}{5}\cos(5x) + C$.
4. $\frac{1}{4}\sin(4t-1) + C$.
5. $\frac{1}{2}\mathrm{e}^{2x+3} + C$.
6. $\mathrm{e} - 1$.
7. 1.
8. Use $u = x^2+1$, $du = 2x\,dx$.
9. Use $u = 3t+2$, $du = 3\,dt$.
10. $-\cos(3x^2 - 2x) + C$.

A.10 Integrals of Sine, Cosine, and Exponentials

1. $-\cos x + C$.
2. $\sin x + C$.
3. $\mathrm{e}^{x} + C$.
4. $-\frac{1}{3}\cos(3t) + C$.
5. $\frac{1}{2}\sin(2x-1) + C$.
6. $\frac{1}{5}\mathrm{e}^{5x} + C$.
7. $-\omega^{-1}\cos(\omega t + p) + C$.
8. 2.
9. 0.
10. $\frac{3}{2}\mathrm{e}^{2t} + 4\cos t + \frac{5}{3}\sin(3t) + C$.

A.11 Integration by Parts

1. Take $u = x$, $dv = \sin x\,dx$: $-x\cos x + \sin x + C$.
2. First obtain $-x^2\cos x + 2\int x\cos x\,dx$. A second integration by parts gives $-x^2\cos x + 2x\sin x + 2\cos x + C$.
3. Two integrations give $I = \mathrm{e}^{x}(\sin x - \cos x) - I$ up to a constant, hence $I = \mathrm{e}^{x}(\sin x - \cos x)/2 + C$.
4. Take $u = x$, $dv = \mathrm{e}^{x}dx$: $(x-1)\mathrm{e}^{x} + C$.
5. $[-x\cos x + \sin x]_{0}^{\pi} = \pi$.
6. Take $u = x$, $v = \sin(2x)/2$: $x\sin(2x)/2 + \cos(2x)/4 + C$.

A.12 Constant-Coefficient First-Order ODEs

1. $y = C\mathrm{e}^{-2t}$.
2. $y = C\mathrm{e}^{3t}$.
3. $y = 5\mathrm{e}^{-4t}$.
4. $y = -\mathrm{e}^{2t}$.
5. $y' = -aC\mathrm{e}^{-at}$, so $y' + ay = 0$.
6. $r + 7 = 0$.
7. $y_p = 3$.
8. $y = 3 + C\mathrm{e}^{-2t}$.
9. $y = 3 - 2\mathrm{e}^{-2t}$.
10. $y_p = -6\mathrm{e}^{2t}$; substitution gives $y_p' - 3y_p = 6\mathrm{e}^{2t}$.

A.13 Constant-Coefficient Second-Order ODEs

1. $r^2 + 5r + 6 = 0$.
2. $y = C_1 \mathrm{e}^{-2t} + C_2 \mathrm{e}^{-3t}$.
3. $y = (C_1 + C_2 t)\mathrm{e}^{2t}$.
4. $y = C_1 \cos(2t) + C_2 \sin(2t)$.
5. $y = \mathrm{e}^{-t}\big(C_1 \cos(2t) + C_2 \sin(2t)\big)$.
6. $y = 3\mathrm{e}^{-2t} - 2\mathrm{e}^{-3t}$.
7. $y = \mathrm{e}^{-2t}\big(C_1 \cos(3t) + C_2 \sin(3t)\big)$.
8. The roots are $-1, -2$, so $y = C_1\mathrm{e}^{-t} + C_2\mathrm{e}^{-2t}$. The data give $C_1 + C_2 = 1$ and $-C_1 - 2C_2 = -1$; hence $y = \mathrm{e}^{-t}$.
9. The roots are $1, -1$, so $y = C_1\mathrm{e}^{t} + C_2\mathrm{e}^{-t}$. The data give $C_1 + C_2 = 0$ and $C_1 - C_2 = 2$; hence $y = \mathrm{e}^{t} - \mathrm{e}^{-t}$.
10. $y = 2 - 2\cos(2t)$.

A.14 Inhomogeneous First-Order Constant-Coefficient ODEs

1. Try a constant: $2y_p = 1$. Thus $y = 1/2 + C\mathrm{e}^{-2t}$.
2. Try $At + B$: $-A = 1$, $A - B = 0$. Thus $y = -t - 1 + C\mathrm{e}^{t}$.
3. Matching coefficients gives $y_p = t^2/2 - t/2 + 1/4$. Thus $y = y_p + C\mathrm{e}^{-2t}$.
4. Integrate directly: $y = t^3/3 + t^2 + t + C$. The initial condition gives $C = 2$.
5. The sine–cosine trial gives $A = 1/5$, $B = -2/5$. Thus $y = (\sin 2t - 2\cos 2t)/5 + C\mathrm{e}^{-t}$.
6. $y_p = (2\sin t - \cos t)/5$. Since $y_p(0) = -1/5$, $C = 1/5$ and $y = (2\sin t - \cos t + \mathrm{e}^{-2t})/5$.

A.15 Inhomogeneous Second-Order Constant-Coefficient ODEs

1. $y_p = 1/2$, so $y = 1/2 + C_1\mathrm{e}^{-t} + C_2\mathrm{e}^{-2t}$.
2. Try $At + B$: $2A = 1$, $3A + 2B = 0$. Thus $y = t/2 - 3/4 + C_1\mathrm{e}^{-t} + C_2\mathrm{e}^{-2t}$.
3. $y_p = t^2/2 - 3t/2 + 7/4$ from coefficient matching; $y = y_p + C_1\mathrm{e}^{-t} + C_2\mathrm{e}^{-2t}$.
4. Two integrations give $y = t^4/12 + C_1 + C_2 t$. The data give $y = t^4/12 + 1 + 2t$.
5. Try $y_p = At^2 + Bt$: $2A = 1$ and $2A + B = 0$. Thus $y = t^2/2 - t + C_1 + C_2\mathrm{e}^{-t}$.
6. $y_p = \sin t/3$ since $y_p'' + 4y_p = \sin t$. Thus $y = \sin t/3 + C_1 \cos 2t + C_2 \sin 2t$.
7. Resonance requires a factor t: $y_p = -t\cos t/2$. Thus $y = -t\cos t/2 + C_1 \cos t + C_2 \sin t$.
8. $y_p = (\sin t - 3\cos t)/10$. Since $y_p(0) = -3/10$ and $y_p'(0) = 1/10$, the constants satisfy $C_1 + C_2 = 3/10$ and $C_1 + 2C_2 = 1/10$. Hence $y = (\sin t - 3\cos t)/10 + \mathrm{e}^{-t}/2 - \mathrm{e}^{-2t}/5$.

A.16 Resonance

1. Here $\omega_0 = 1$, so $y_p = -t\cos t/2$. The general solution is $y = -t\cos t/2 + C_1\cos t + C_2\sin t$.

2. Here $\omega_0 = 3$, so $y_p = -t\cos(3t)/6$. The general solution is $y = -t\cos(3t)/6 + C_1\cos(3t) + C_2\sin(3t)$.

3. A particular solution is $y_p = -t\cos(2t)/4$, so the general solution is $y = -t\cos(2t)/4 + C_1\cos(2t) + C_2\sin(2t)$. The data give $C_1 = 1$ and $-1/4 + 2C_2 = 0$, hence $y = -t\cos(2t)/4 + \cos(2t) + \sin(2t)/8$.

4. Resonance occurs at $\omega = 4$, the homogeneous angular frequency. A particular solution is $y_p = -t\cos(4t)/8$. The general solution is

$$y = -t\cos(4t)/8 + C_1\cos(4t) + C_2\sin(4t).$$

5. The first equation is nonresonant: $y_p = \sin(2t)/(9-4) = \sin(2t)/5$, so $y = \sin(2t)/5 + C_1\cos(3t) + C_2\sin(3t)$. Every such solution is bounded because it is a finite sum of bounded sines and cosines. The second equation is resonant: $y_p = -t\cos(3t)/6$, so $y = -t\cos(3t)/6 + C_1\cos(3t) + C_2\sin(3t)$. Its growing term cannot be canceled by the bounded homogeneous terms; the oscillations are unbounded in amplitude.